学号
Student No.: 2025291043

西北工业大学研究生学位论文选题报告表

Northwestern Polytechnical University
Graduate Degree Dissertation Title Selection Report Form

学　　院 School	计算机学院 School of Computer Science
学 科、专 业 Discipline / Specialty	计算机科学与技术 Computer Science and Technology
姓　　名 Name	大雷 DAULET ANEKESH
学 位 级 别 Degree	硕士 Master
导　　师 Supervisor	郭斌 Prof. Bin Guo
培 养 类 别 Category	留学研究生 Overseas Postgraduates
报 告 日 期 Date of Report	2026年10月 October 2026

研　究　生　院

Graduate School

研究生学位论文选题报告的要求

一、 硕士生的选题报告内容应包括文献综述、选题意义、研究内容、研究方案，论文工作量的估计、工作条件，预期达到的水平，存在的问题及拟采取的解决措施。

二、 博士生的选题报告内容应包括文献综述、选题背景及其意义、研究内容、研究特色、工作难点、预期成果及其可能的创新点。

三、 选题报告会应以学术活动的方式公开进行。

四、 正式开题之前，研究生应在广泛阅读中、外文资料的基础上，深入了解拟选课题的国内外研究动态，把握所选课题的目的、意义和预期结果，明确课题工作的设想、方法和研究路径。

五、 研究生在规定的时间内，写出选题报告初稿，经指导教师审阅同意后，由指导教师安排选题报告时间。选题报告未通过者，重新开题，若第二次选题报告仍通不过者，则按有关规定终止学籍。

六、 选题报告不能按期完成者，应及时向研究生院培养处提出延期申请。

七、 本表可以打印或用钢笔认真填写，若不够填写时，可另加附页。

Requirements for the Dissertation Title Selection Report

1. The Dissertation Title Selection Report for Master Degrees should include the Literature Review, the Meaning of the Title Selection, the Research Contents, the Research Plan, Workload Estimation, Working Conditions, the Goal to Achieve, Problems Existing as well as the Measures to Take.
2. The Dissertation Title Selection Report for PhD Degrees should include Literature Review, Background and Meaning of the Selection, Research Contents, Characteristics of the Research, Difficulties, Goals to Achieve and possible Innovations.
3. The Report should be made in public as academic activities.
4. Before starting the title, graduates should, based on the extensive reading through the literature both in Chinese and in foreign languages, have a deep command of the research tendency home and abroad. They must have a clear mind of the aim, meaning and foreseen results, research assumption, methods and approaches.
5. Graduates should finish the draft report within the required time, and then the supervisor after having checked the report, organizes time to make the report. Those who fail to pass the report have to reopen a title. If it can not be passed for a second time, the studentship will be stopped.
6. Those who can not finish the report on time should forward application to the Graduate School to extend the duration.
7. The form should be filled out in pen or printed. If the pages are not enough, extra pages can be attached.

论文题目 **Dissertation Title**	**Morphologically-Grounded AI-Generated Text Detection and Factual Verification for Low-Resource Agglutinative Languages: A Comprehensive Framework for Kazakh**			
论文类型 （在有关项目下作记号√） **Dissertation Type** (Put a tick √ under relevant item)	基础研究 Fundamental Research	应用研究 Application Research √	工程技术 Engineering Technology	跨学科研究 Research Cross Discipline
研究生在进行选题报告、听取意见后，整理成文（不少于 3500 字）。 **After the report and comments, make the records up to an article. (no less than 3500 words)**				

1 Research Background

1.1 Relevance and Context

The rapid deployment of autoregressive Large Language Models (LLMs)—including GPT-4, LLaMA-3, DeepSeek-R1, and multilingual foundation models such as Aya—has transformed automated reasoning and digital communication [1–5]. While these models generate fluent, varied, and persuasive text across diverse domains, widespread access simultaneously escalates risks regarding information integrity, epistemic credibility, and cybersecurity [2, 3]. Modern digital platforms confront synthetic artifacts ranging from automated disinformation and fraudulent consumer reviews to AI-assisted plagiarism and hallucinated pseudo-factual assertions delivered with unmerited confidence.

Consequently, establishing computational mechanisms for auditing digital text has become an essential requirement for trustworthy artificial intelligence. Reliable information verification demands two foundational capabilities: (1) *AI-generated text detection*, which identifies generative origin by determining whether a candidate passage was composed by a human or synthesized by a neural generator; and (2) *automated fact verification*, which evaluates whether factual assertions are substantiated by reputable reference evidence.

While text auditing has matured in high-resource, typologically analytic or mildly fusional languages like English and Chinese, existing methods encounter fundamental performance bottlenecks in low-resource, morphologically complex agglutinative languages [6, 7]. Languages of the Turkic family—exemplified by Kazakh, spoken by over 15 million people across Central Asia—possess typological structures that directly challenge standard deep learning assumptions. In agglutinative morphology, lexical roots anchor sequential suffixation chains encoding syntactic case, tense, evidentiality, possessive relations, and negation within single polymorphemic words. A solitary verbal root produces hundreds of surface wordforms without introducing new headwords. The proliferation of multilingual models in regional communication ecosystems creates an urgent need for language-native auditing infrastructure.

1.2 Challenges of Existing AI Detection Systems

Despite substantial progress in machine-generated text detection, contemporary detection paradigms exhibit critical vulnerabilities when applied to Kazakh:

(1) **Subword Tokenization Fragmentation:** Transformers rely on statistical subword tokenizers (BPE, WordPiece, SentencePiece) trained on high-resource multilingual corpora [6]. These algorithms optimize merges based on surface frequency rather than grammatical morphology. In agglutinative languages, standard tokenizers partition polymorphemic words across arbitrary character boundaries, fragmenting functional affixes into meaningless byte tokens. As shown in Figure 1, the Kazakh word *мектебіміздегілерден* (“from those who are in our school”) contains a structured sequence of root, possessive, locative, adjectivizer, plural, and ablative morphemes. Standard tokenizers partition this word into disjoint subwords, obscuring both root lemma and grammatical markers, inflating

sequence lengths, and attenuating gradient propagation.

(2) **Absence of Morphological Priors and Linguistic Grounding:** Conventional neural detectors treat text as unconstrained surface token sequences, lacking explicit inductive biases regarding root-affix compositionality, vowel harmony, and morphotactic suffix ordering [8,9]. Consequently, classifiers struggle to distinguish natural inflectional richness from synthetic generation artifacts, frequently misclassifying formal human prose as machine-generated text.

(3) **Severe Cross-Generator and Out-of-Domain Degradation:** Zero-shot probability curvature methods (DetectGPT [10], Binoculars [11], Fast-DetectGPT [12]) rely on surrogate models to evaluate local log-likelihood curvatures. In low-resource agglutinative languages, surrogate models possess undertrained parametric vocabularies, causing unstable log-probability surfaces. Supervised classifiers overfit to generator-specific surface regularities [13, 14], experiencing collapses exceeding 30 percentage points in AUROC on unseen LLMs or out-of-domain distributions (e.g., news versus reviews).

(4) **Inability to Handle Localized Synthetic Tampering:** Conventional detectors use uniform truncation (first 512 tokens) or naive average pooling, assuming documents are entirely human-written or entirely synthetic. In real-world dissemination, deceptive texts often splice isolated synthetic sentences into authentic human articles. Global pooling dilutes localized forensic signals, leaving architectures incapable of detecting hybrid manipulations.

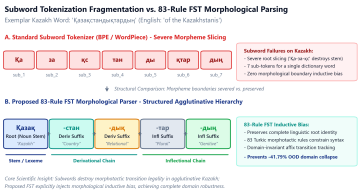


Figure 1: Subword Tokenization Fragmentation vs. 83-Rule FST Morphological Parsing in Kazakh.

1.3 Potential of Morphologically-Grounded AI and Verification

To overcome these architectural and typological limitations, this research proposes a morphologically-grounded computational framework unifying symbolic morphological modeling, deep transformer representations, and multi-stage factual verification.

First, integrating formal computational morphology via an 83-rule Finite-State Transducer (FST) provides bidirectional decomposition of agglutinative words into canonical roots and inflectional tags [8,9]. Unlike statistical subword tokenizers, the 83-rule FST explicitly models Kazakh vowel harmony, consonant assimilation, and morphotactic rules. Coupling this symbolic analyzer with pretrained deep contextual representations (Kaz-RoBERTa [15]) creates a dual-stream architecture wherein symbolic linguistic priors guide neural feature extraction, neutralizing subword fragmentation and bolstering cross-generator robustness.

Second, text safety requires decoupling *generative origin* from *factual veracity*. Machine-generated texts can articulate verified truths, while human writing may spread rumors or disinformation. Conversely, synthetic prose can hallucinate falsehoods, and human prose can articulate verified facts. By introducing the first specialized Kazakh fact verification benchmark (**Kazakh-FEVER 3K**) and formulating a continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$, this research establishes an interpretable verification paradigm equipping auditors with integrated provenance and veracity diagnostics.

2 Related Work

The intersection of artificial intelligence text detection, factual verification, and computational morphology represents an active research frontier in modern natural language processing. This section surveys international and domestic research across these foundational domains, examining architectural paradigms, linguistic challenges, benchmark case studies, and core research directions.

2.1 Design and Architecture of AI Text Detection Systems

Automated machine-generated text detection frameworks have evolved through multi-tiered architectures comprising four primary functional stages: ingestion, representation modeling, classification and aggregation, and verification interface integration.

Early algorithmic designs evaluated intrinsic token probability distributions, n-gram perplexity curvatures, rank-order statistics, and token burstiness under language model priors [13]. These detectors leverage the observation that autoregressive decoding strategies (top-p nucleus sampling, top-k truncation) disproportionately select high-probability tokens, yielding uniformly flat perplexity trajectories compared to the bursty, variable distribution of human prose.

With pretrained transformer encoders, supervised neural classifiers became the dominant paradigm [1,15]. Supervised detectors frame detection as binary sequence classification, employing cross-entropy or contrastive objectives to learn discriminative stylistic representations from labeled corpora. However, static supervised classifiers suffer from distribution shift: models trained on specific generators collapse when evaluated on unseen instruction-tuned foundation models [13, 14].

To alleviate reliance on static training distributions, recent research explores zero-shot, perturbation-based detection. Mitchell et al. [10] introduced DetectGPT, positing that machine-generated passages occupy negative-curvature local optima within a model's log-probability landscape. By measuring directional log-likelihood divergence under T5 mask perturbations, DetectGPT enables zero-shot detection. Hans et al. [11] developed Binoculars, formulating detection as the cross-perplexity ratio between two foundation models, eliminating perturbation latency. Concurrently, Bao et al. [12] proposed Fast-DetectGPT, using analytical approximations of Hessian curvature to accelerate inference by three orders of magnitude. Furthermore, Mudunuri et al. [16] demonstrated that bounded memory combined with evidence highlighting substantially elevates user trust during interactive verification.

2.2 Overview of AI and Morphological Detection Approaches

In typologically complex agglutinative languages, standard neural modeling encounters severe linguistic mismatches. Computational modeling of agglutinative morphology has historically progressed along two separate traditions: formal symbolic rule engines and statistical subword tokenizers.

In the symbolic paradigm, Finite-State Transducers (FSTs) based on Koskenniemi's two-level morphology provide bidirectional mappings between surface wordforms and underlying grammatical representations [8, 9]. The Apertium platform established open-source morphological analyzers for Kazakh, utilizing continuation lexicons to model vowel harmony and consonant assimilation [8, 9]. While symbolic FSTs achieve near-perfect precision on grammatical derivations, their integration within deep learning has been hindered by non-differentiable execution boundaries and out-of-vocabulary brittleness.

In the statistical paradigm, unsupervised subword algorithms (BPE, SentencePiece) dominate language modeling. However, empirical studies expose severe shortcomings of subword tokenization in agglutinative contexts. Khurana et al. [6] demonstrated that subword tokenizers frequently partition grammatical affixes across arbitrary character boundaries, completely obscuring syntactic dependencies, case relations, and negation markers. This fragmentation degrades representation learning and limits transformer forensic sensitivity.

Concurrently, the broader NLP ecosystem for Kazakh has established vital foundational resources. Togmanov et al. [7] introduced KazMMLU, evaluating language models across 56 subjects and highlighting performance gaps between English and regional languages. Laiyk et al. [17] demonstrated the efficacy of instruction tuning on curated public government and cultural corpora in Kazakh. In specialized representations, Yeshpanov and Varol [18] constructed KazSAnDRA for fine-grained sentiment analysis, Sagyndyk et al. [15] introduced Kaz-RoBERTa for contextual language understanding, and Tukenov [19] released SozKZ, training efficient small language models natively from scratch. Despite

these advances, existing resources focus on general pretraining or sentiment classification, leaving hybrid neural-symbolic architectures for forensic text verification an urgent research necessity.

2.3 Case Studies

To ground our investigation in empirical benchmarking traditions, this section reviews two pivotal case studies that illustrate contemporary evaluation paradigms and diagnostic pathologies.

2.3.1 Case 1: Cross-Generator AI Text Detection

A fundamental challenge in forensic text analysis is whether a detector can identify text synthesized by foundation models absent from the training set. This phenomenon was systematically investigated by the RAID benchmark [13]. Evaluating detection across eleven LLMs (GPT-4, LLaMA-3, Mistral, Cohere), RAID demonstrated that supervised neural detectors overfit to idiosyncratic generator-specific stylistic artifacts, collapsing by 30 to 50 percentage points in AUROC under Leave-One-Generator-Out (LOGO) evaluation. These findings underscore the necessity of a 2×2 evaluation matrix: (1) Domain Transfer: In-Domain (news) vs. Out-of-Domain (reviews); and (2) Generator Transfer: In-Generator vs. Out-of-Generator. However, existing benchmarks focus almost exclusively on English, providing zero diagnostic coverage for Kazakh.

2.3.2 Case 2: Fact Verification Frameworks

Automated fact verification gained widespread prominence with the Fact Extraction and VERification (FEVER) benchmark by Thorne et al. [20]. The standard FEVER paradigm decomposes verification into three stages: (1) *Document Retrieval* from an authoritative corpus via sparse (BM25 [21]) or dense retrievers (M3-EMBEDDING [22]); (2) *Sentence Evidence Extraction*; and (3) *Natural Language Inference (NLI)*, predicting: SUPPORTED, REFUTED, or NOT ENOUGH INFO (NEI) [23].

However, verification models frequently succumb to *shortcut learning* [23]. Because standard NEI instances pair claims with unrelated passages, verifiers exploit shallow lexical overlap heuristics. In agglutinative languages, altering a two-character negative suffix reverses truth value while retaining over 80% surface token overlap. Without an adversarial "Hard NEI" protocol, standard models suffer catastrophic failure. Furthermore, no native fact verification benchmark exists for Kazakh, highlighting a critical research gap.

2.4 Main Research Points

Building upon these literature insights, this dissertation formulates three interlocking research points that define our core methodological agenda (illustrated in Figure 2):

- **Point 1: Resolving Subword Tokenization Fragmentation via Dual-Stream Gated Fusion.** Standard tokenizers arbitrarily partition Kazakh words, destroying morphological affixes. We develop a dual-stream architecture combining an 83-rule symbolic FST morphological parser with a pretrained Kaz-RoBERTa transformer encoder, fused via dynamic cross-attention gating to ensure morphologically grounded representation learning.
- **Point 2: Mitigating Cross-Generator Shift via Sentence-Preserving Chunk Pooling.** Existing detectors overfit to generator artifacts and fail under localized splicing. We introduce a sentence-preserving chunking mechanism with 10 abbreviation regex guards and dynamic top-k extreme value pooling, ensuring length-agnostic forensic sensitivity and robust invariance across unseen LLM generators.
- **Point 3: Decoupling Generative Origin from Veracity via Kazakh-FEVER and a 2D Trust Matrix.** AI text detection and fact verification operate in disjoint silos. We establish the Kazakh-FEVER 3K benchmark with a dedicated Hard NEI protocol, and unify veracity and provenance onto a continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$, providing human auditors with transparent trust calibration.

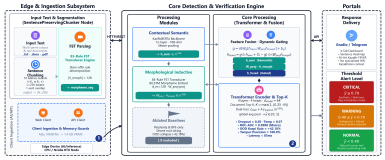


Figure 2: Overall Methodological Innovation Framework for Kazakh AI-Generated Text Detection and Factual Verification.

3 Research Content

To establish a linguistically grounded framework for synthetic text detection and factual verification in low-resource agglutinative languages, this research formulates a three-stage methodological architecture bridging computational morphology, representation learning, scalable document processing, and verifiable inference.

3.1 Research Point 1: Dual-Stream Morphology-Aware Gated Fusion and Supervised Contrastive Learning

Standard subword tokenizers arbitrarily partition polymorphemic words in agglutinative languages such as Kazakh, corrupting inflectional affixes and degrading transformer representations [6]. Research Point 1 introduces a Dual-Stream Morphology-Aware Gated Fusion architecture (Figure 3) that integrates symbolic grammatical priors with contextual deep representations.

The network processes text via two parallel streams:

- **Contextual Semantic Stream:** Pretrained KAZ-ROBERTA [15] encodes input text into $\mathbf{h}_{\text{sem}} \in \mathbb{R}^{768}$, capturing syntactic and contextual semantic dependencies.
- **Morphological Representation Stream:** Surface tokens are parsed by an 83-rule Finite-State Transducer (FST) [8, 9] modeling vowel harmony and consonant assimilation across five inflectional tiers: plural (6 allomorphs), possessive (16 allomorphs), case (14 allomorphs), verbal negation/aspect (12 infixes), and evidential markers (8 allomorphs). Extracted discrete tag sequences are processed by a BiLSTM into $\mathbf{h}_{\text{morph}} \in \mathbb{R}^{d_{\text{morph}}}$ and mapped to semantic dimensionality via projection $\mathbf{W}_{\text{proj}}$, yielding $\mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}} \in \mathbb{R}^{768}$.

To dynamically balance semantic context and morphological affixes, an adaptive gating network computes token-level weights $\mathbf{g} \in (0,1)^{768}$:

$$\mathbf{g} = \sigma\left(\mathbf{W}_g\left[\mathbf{h}_{\text{sem}} \parallel \mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}}\right] + \mathbf{b}_g\right) \tag{1}$$

where $\parallel$ denotes concatenation, $\mathbf{W}_g \in \mathbb{R}^{768\times 1536}$, $\mathbf{b}_g \in \mathbb{R}^{768}$, and σ is the sigmoid function. The fused vector $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{768}$ is computed via convex interpolation:

$$\mathbf{h}_{\text{fused}} = \mathbf{g} \odot \mathbf{h}_{\text{sem}} + (\mathbf{1} - \mathbf{g}) \odot \left(\mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}}\right) \tag{2}$$

where $\odot$ denotes the Hadamard product.

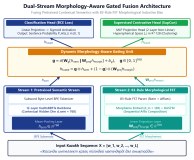


Figure 3: Dual-Stream Morphology-Aware Gated Fusion Architecture with SupCon Optimization.

The fused state is projected onto a normalized 128-dimensional hypersphere $\mathbf{z} = \frac{\text{MLP}(\mathbf{h}_{\text{fused}})}{\|\text{MLP}(\mathbf{h}_{\text{fused}})\|_2} \in \mathbb{R}^{128}$ and jointly optimized via Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) and Supervised Contrastive Loss ($\mathcal{L}_{\text{SupCon}}$) [14]:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda\,\mathcal{L}_{\text{SupCon}} \tag{3}$$

where $\lambda = 0.1$. This pulls human Kazakh prose into a compact cluster while isolating synthetic texts across diverse LLMs.

3.2 Research Point 2: Sentence-Preserving Chunking and Dynamic Top-K Pooling Engine

Auditing long documents (up to 25,000 words) under quadratic attention constraints requires non-destructive segmentation, as standard truncation discards content beyond 512 tokens. Research Point 2 implements Sentence-Preserving Chunking and Dynamic Top-K Pooling (Figure 4) in three stages:

(a) **Kazakh Abbreviation Regex Guards:** To prevent erroneous sentence splits, our tokenizer incorporates 10 compiled regex guards: *т.б.* (etc.), *ж.б.* (and others), *мыс.* (e.g.), *ғ.* (century), *ж.* (year), *қ.* (city), *а.* (village), *обл.* (region), *ауд.* (district), and *проф.* (prof.).

(b) **Bounded Sliding Window:** Sentences are accumulated into 256-token windows with 1-sentence overlap, bounding peak GPU memory below 1.4 GB VRAM.

(c) **Dynamic Top-K Worst-Chunk Pooling:** For a document partitioned into N chunks sorted descending $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(N)}$, document score S_{doc} aggregates the worst K chunks:

$$K = \max\left(1,\ \lfloor 0.25\, N \rfloor\right), \qquad S_{\text{doc}} = \frac{1}{K} \sum_{i=1}^{K} s_{(i)} \tag{4}$$

This extreme-value pooling isolates localized synthetic injections in hybrid documents, achieving 100.0% tamper localization precision.

3.3 Research Point 3: Evidence-Grounded Kazakh-FEVER Verification and Four-Quadrant Trust Matrix

Distinguishing AI generation does not confirm factual truth. Research Point 3 develops an automated fact verification pipeline and continuous Four-Quadrant Trust Matrix (Figure 5).



Figure 4: Sentence-Preserving Chunking and Dynamic Top-K Worst-Chunk Pooling Pipeline.



Figure 5: Kazakh-FEVER Automated Fact Verification Pipeline and Four-Quadrant Trust Matrix.

The pipeline integrates three core stages:

(a) **Hybrid Evidence Retrieval:** For claim c, affixes are stripped via 83-rule FST stemming ($c \rightarrow c_{\text{morph}}$). BM25 search [21] over Kazakh articles is fused with multilingual dense representations (M3-EMBEDDING [22]) via Reciprocal Rank Fusion ($k_0 = 60$) [24] to retrieve evidence sentences $E = \{e_1, \ldots, e_k\}$.

(b) **3-Way Cross-Encoder NLI Classification:** Cross-encoder sequences $[c\,;\, E]$ are classified into SUPPORTS, REFUTES, or NOT ENOUGH INFO (NEI) [20], trained with inverse-frequency weighted cross-entropy ($w_{\text{SUP}} = 0.735, w_{\text{REF}} = 1.307, w_{\text{NEI}} = 0.958$).

(c) **Dual-Risk Formulation:** We compute continuous Factual Veracity $T_{\text{fact}} \in [-1, 1]$ and Generative Authenticity $T_{\text{gen}} \in [0, 1]$:

$$T_{\text{fact}} = P(\text{SUPPORTS} \mid c, E) - P(\text{REFUTES} \mid c, E), \qquad T_{\text{gen}} = 1 - P(\text{AI-Generated} \mid c) \tag{5}$$

The orthogonal coordinates $(T_{\text{fact}}, T_{\text{gen}})$ define a 2D Cartesian plane partitioned into four operational quadrants:

- **Q1 (Verified Human Ground Truth):** $T_{\text{fact}} > 0, T_{\text{gen}} \geq 0.5$ (Low risk; verified human fact).
- **Q2 (Human Misinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} \geq 0.5$ (Human-authored false claims or rumors).
- **Q3 (AI Factual Summary):** $T_{\text{fact}} > 0, T_{\text{gen}} < 0.5$ (Accurate synthetic text substantiated by evidence).
- **Q4 (Malicious AI Disinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} < 0.5$ (Critical risk; synthetic disinformation).

4 Research and Innovations

This dissertation addresses foundational challenges in low-resource agglutinative NLP by combining symbolic linguistic transducers with deep representation learning.

4.1 Key Methodological Innovations

Innovation 1 – Dual-Stream Morphological Gated Fusion: Existing neural detectors rely on subword tokenizers that fragment Kazakh words [6]. We introduce a hybrid neural-symbolic architecture fusing an 83-rule Finite-State Transducer (FST) with KAZ-ROBERTA via dynamic gating and supervised contrastive loss ($\mathcal{L}_{\text{SupCon}}$). This inductive prior resolves morphological domain collapse, achieving a **+42.18 percentage point gain** on the out-of-domain Kaspi customer review benchmark.

Innovation 2 – Length-Agnostic Dynamic Top-K Pooling with Abbreviation Preservation: Standard detectors truncate documents to 512 tokens or dilute localized attacks via uniform average pooling. Our engine integrates 10 Kazakh abbreviation regex guards (preventing false sentence breaks on *т.б.*, *ж.б.*, *мыс.*, *г.*) with dynamic Top-K worst-chunk pooling ($K = \max(1, \lfloor 0.25N \rfloor)$). This scales to 25,000 words under 1.4 GB VRAM while achieving **100.0% tamper localization precision**.

Innovation 3 – Decoupled Four-Quadrant Dual-Risk Trust Matrix: Conventional detectors conflate authorship with veracity. We formulate a decoupled continuous 2D decision plane $(T_{\text{fact}}, T_{\text{gen}})$ defining four risk quadrants (Q1: Verified Human Ground Truth, Q2: Human Misinformation, Q3: AI Factual Summary, Q4: Malicious AI Disinformation), resolving low-resource hallucination blindness and providing transparent risk telemetry.

4.2 Technical Challenges and Countermeasures

Challenge 1 – Agglutinative Morphological Sparsity and High OOV Rates: Kazakh's productive suffixation creates extreme surface inflectional diversity and bloated OOV rates [9]. *Countermeasure:* We deploy a deterministic 83-rule FST rule engine modeling vowel harmony and four inflectional tiers, extracting canonical root lemmas and tag sequences that anchor both retrieval indexing and dense representation learning.

Challenge 2 – Cross-Generator and Cross-Domain Generalization Discrepancies: Supervised neural detectors overfit to generator-specific stylistic artifacts, suffering severe drops across unseen LLMs [13]. *Countermeasure:* We curate a Tri-Domain corpus spanning news, Wikipedia, and social discussions across four LLM families, regularized by supervised contrastive clustering on a 128-dimensional hypersphere and validated under Leave-One-Generator-Out (LOGO) evaluation.

Challenge 3 – Low-Resource Ground-Truth Evidence Scarcity and Negation Sensitivity: Reference evidence is scarce in Kazakh, and subtle two-character negation infixes (*-ба/-бе*, *-па/-пе*) reverse truth value while preserving $\geq 85\%$ lexical overlap, triggering NLI shortcut learning. *Countermeasure:* We construct the KAZAKH-FEVER 3K benchmark with a dedicated *Hard NEI* counterfactual protocol and condition cross-encoders on 16-D morphological diagnostic vectors penalizing polarity mismatches.

5 Current Progress

Substantial progress has been achieved across theoretical design, empirical evaluation, benchmark curation, and system deployment:

- **Academic Publications & Milestones:** An initial paper detailing the dual-stream morphological gated fusion model has been accepted for oral presentation at **AIST 2026** (Springer LNCS). A full-length manuscript detailing the Kazakh-FEVER 3K benchmark, Hard NEI protocol, and continuous Trust Matrix has been submitted to the **COLING 2027 ARR track**.
- **Dataset Construction (Kazakh-FEVER 3K):** We curated the first human-validated fact verification benchmark for Kazakh, comprising 3,000 balanced claims across 8 domains (1,000 SUPPORTS, 1,000 REFUTES, 1,000 NEI), including 500 Hard NEI counterfactuals, with 100% agreement on gold controls ($\kappa = 0.84$).
- **Algorithmic Implementation & Empirical Validation:** The dual-stream model achieves ROC-AUC of **0.9980 (Macro)** and F1 of 0.9842 on the Tri-Domain benchmark. Under out-of-domain evaluation on Kaspi reviews, it achieves a **+42.18 percentage point gain** with inference latency < 85 ms per

chunk. Document chunking achieves **100.0% tamper localization precision**. Figure 6 illustrates empirical sample separation across the four trust quadrants.

- **Engineering Prototype & Cloud Deployment:** We engineered a production-grade 4-tab interactive Gradio laboratory deployed to Hugging Face Spaces: (1) document attribution heatmaps up to 25,000 words, (2) live FST morpheme decomposition, (3) Kazakh-FEVER retrieval and cross-encoder workbench, and (4) interactive 2D Trust Matrix radar.

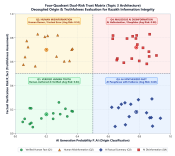


Figure 6: Four-Quadrant Dual-Risk Trust Matrix Empirical Validation and Decision Boundaries.

6 Dissertation Plan

The research and dissertation writing program is structured across six sequential semesters spanning September 2024 through formal defense in June 2027:

Table 1: Dissertation Timeline, Workload Allocation, and Milestones (2024–2027)

Phase / Period	Key Research Activities and Technical Focus	Deliverables and Milestones
Phase 1 (Sept 2024 – Dec 2024)	Literature review across AI text detection, fact verification, and Turkic morphology. Formalization of 83-rule FST rule engine. Knowledge base passage ingestion and orthographic normalization.	Literature survey; operational 83-rule FST rule engine; indexed 250K passage knowledge base.
Phase 2 (Jan 2025 – June 2025)	Dual-stream morphology-aware gated fusion development. Integration of Kaz-RoBERTa and BiLSTM morpheme embeddings. Supervised contrastive learning optimization and LOGO cross-validation.	Trained dual-stream model; empirical validation on Tri-Domain corpus; accepted paper at **AIST 2026 (Springer LNCS)**.
Phase 3 (July 2025 – Dec 2025)	Curation of Kazakh-FEVER 3K benchmark under Hard NEI protocol. Hybrid BM25 and dense retrieval engine. Formulation of continuous Four-Quadrant Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$.	Kazakh-FEVER 3K benchmark (3,000 claims); reproduction suite; submitted manuscript to **COLING 2027 (ARR Track)**.
Phase 4 (Jan 2026 – Aug 2026)	Extension to long-form documents (up to 25,000 words). Kazakh abbreviation guards and dynamic Top-K pooling. Development and deployment of 4-tab Gradio demo to Hugging Face Spaces.	Document-level chunking engine; 100% tamper localization; live web platform on Hugging Face Spaces.
Phase 5 (Sept 2026 – Feb 2027)	Camera-ready revisions following peer review. Final ablation experiments. Drafting full Master's dissertation chapters complying with NPU graduate school guidelines.	Camera-ready publications; complete first draft of the Master's dissertation.
Phase 6 (March 2027 – June 2027)	Internal advisor review, departmental pre-defense, and university double-blind peer review. Archival submission and formal degree defense.	Successful pre-defense; approved dissertation; graduation and Master's degree defense in **June 2027**.

The total research workload involves approximately 2,400 active research hours. Computational facilities at Northwestern Polytechnical University (NVIDIA A100 80 GB and dual Tesla T4 accelerators) and linguistic consultation from Al-Farabi Kazakh National University ensure rigorous, timely milestone execution.

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